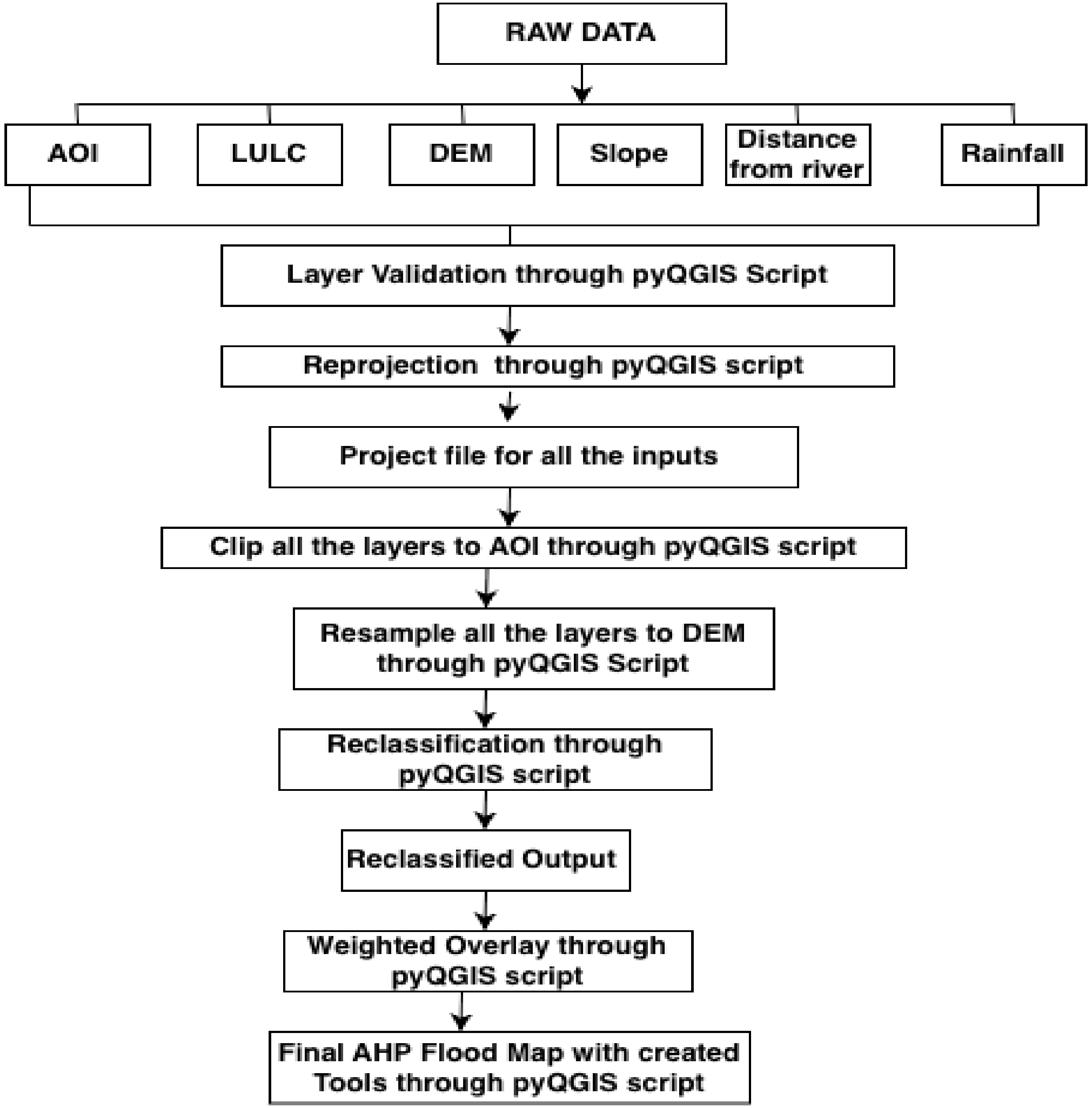
Flood Susceptibility Analysis using PyQGIS and Flood Tool Creation in QGIS

Introduction

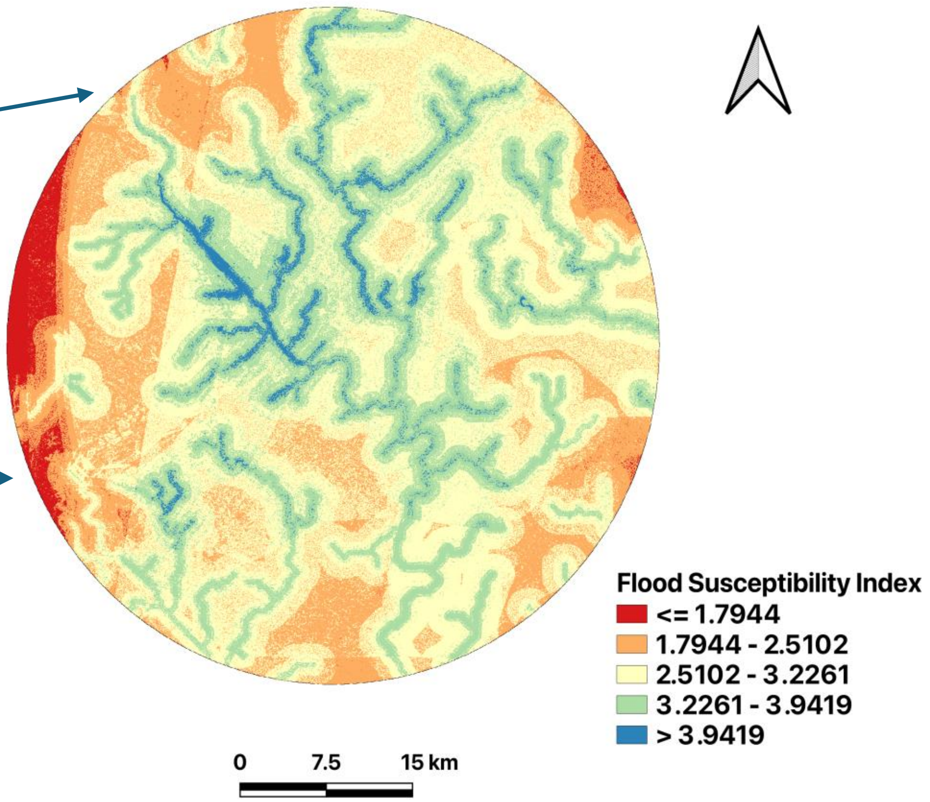
- Floods are one of the most frequent and damaging natural disasters, causing significant environmental, economic, and social impacts.
- Accurate flood susceptibility mapping is essential for disaster management, land-use planning, and risk reduction.
- This study uses **Remote Sensing, GIS, and Analytical Hierarchy Process (AHP)** techniques to prepare a flood susceptibility map.
- Multiple thematic layers such as **DEM, Slope, Rainfall, LULC, and Distance from River** are integrated for analysis.
- Automated **PyQGIS scripts** are used for preprocessing, validation, reprojection, clipping, resampling, reclassification, and weighted overlay analysis.
- The workflow improves accuracy, reduces manual effort, and creates a reproducible flood mapping methodology.
- The final output is an **AHP-based Flood Susceptibility Map** identifying areas with varying flood risk levels.

Methodology

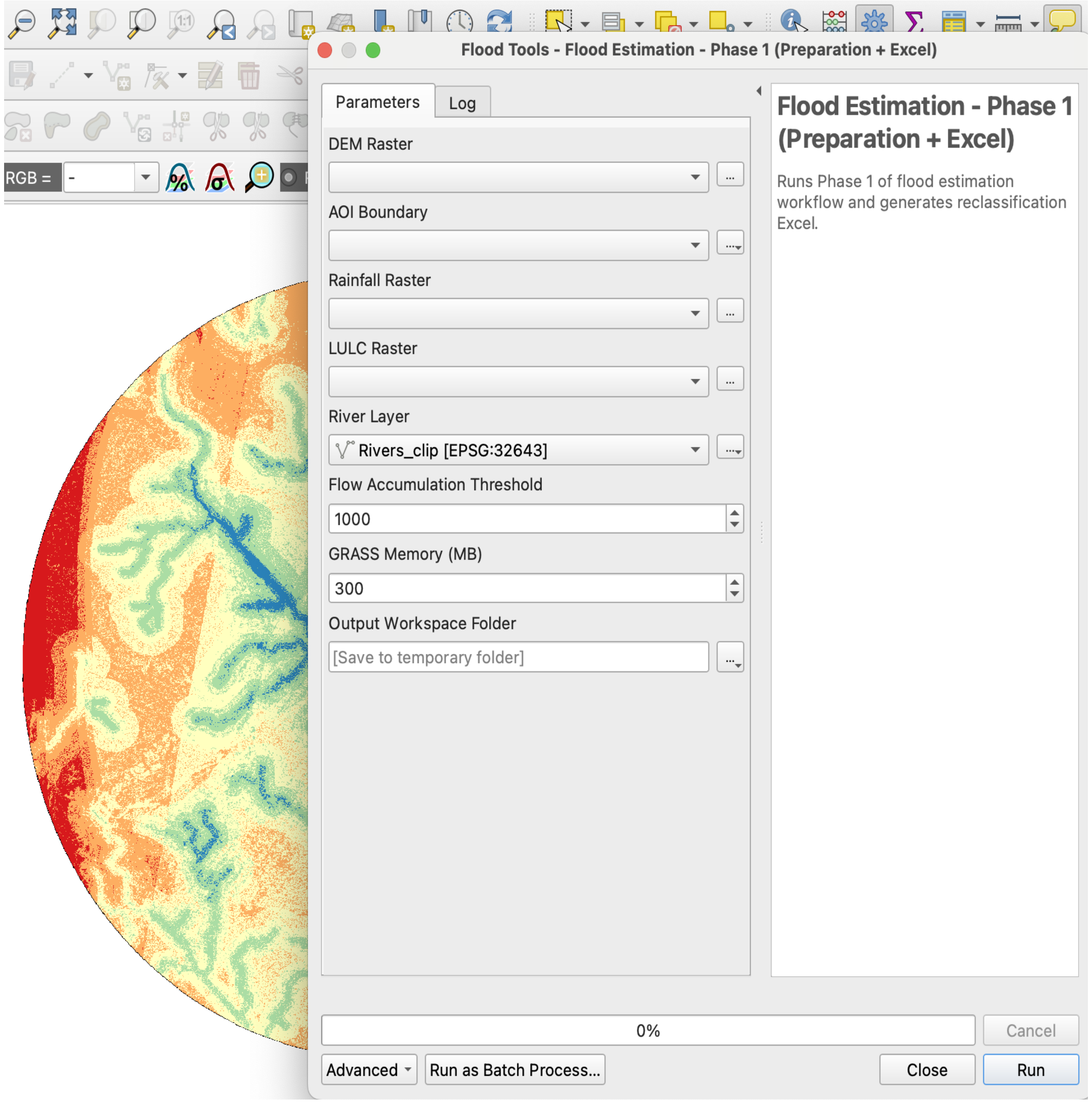


Generated maps and tools through PyQGIS

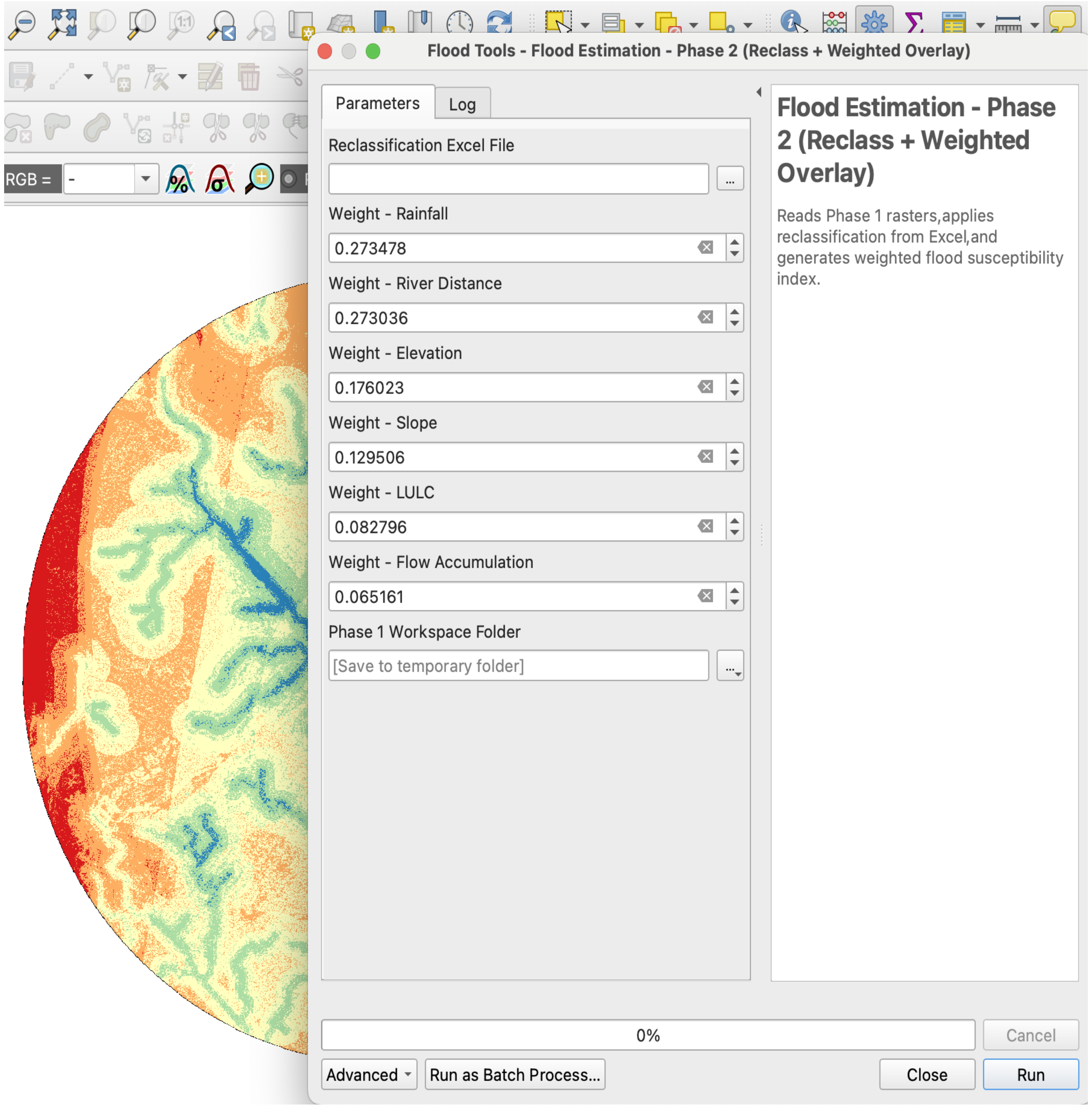
- Scripts
  - Flood Tools
    - Flood Estimation - Phase 1 (Preparation + Excel)
    - Flood Estimation - Phase 2 (Reclass + Weighted Overlay)
    - Flood Estimation Custom Reclass



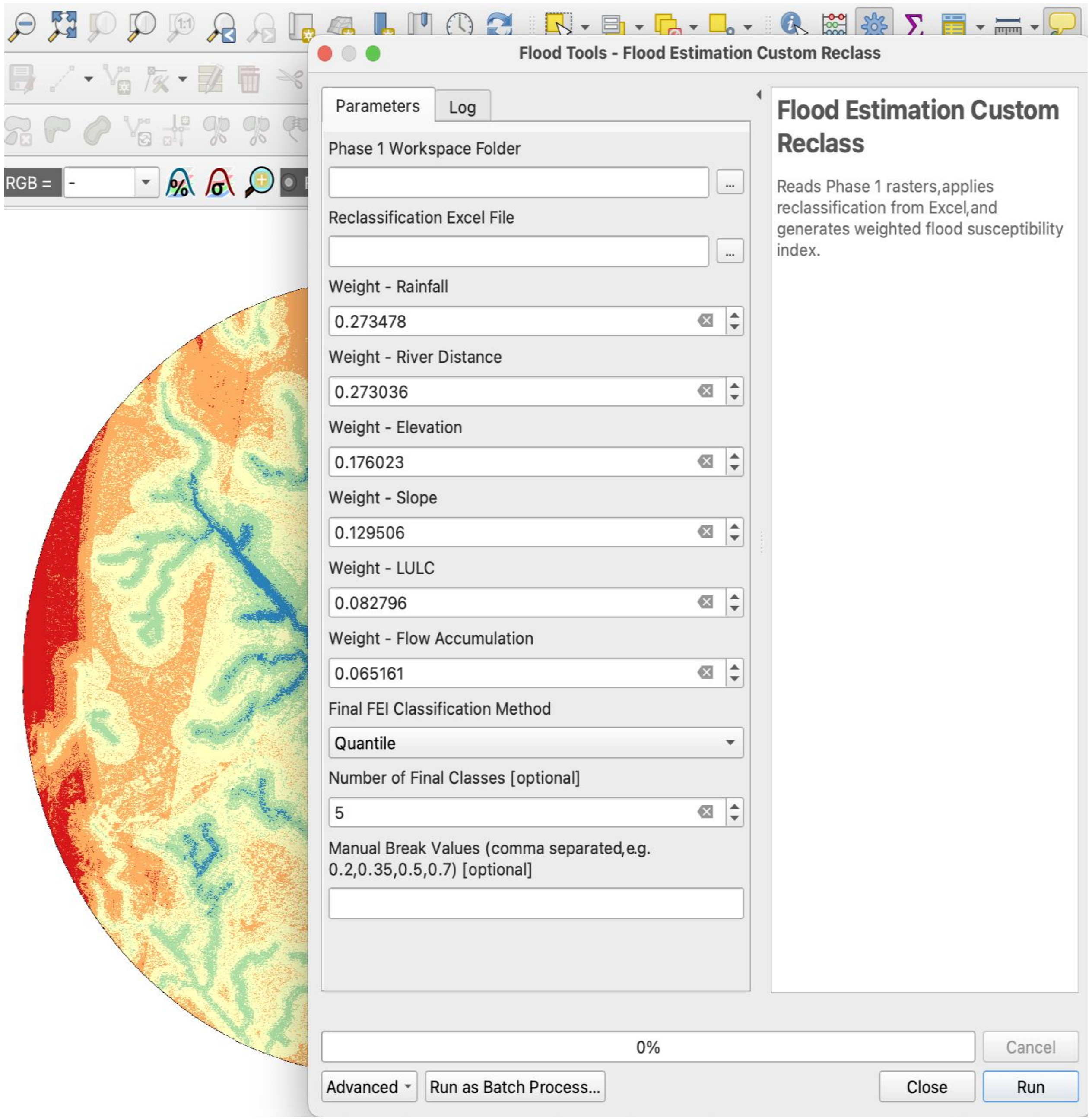
Tool created for Flood Estimation Parameters

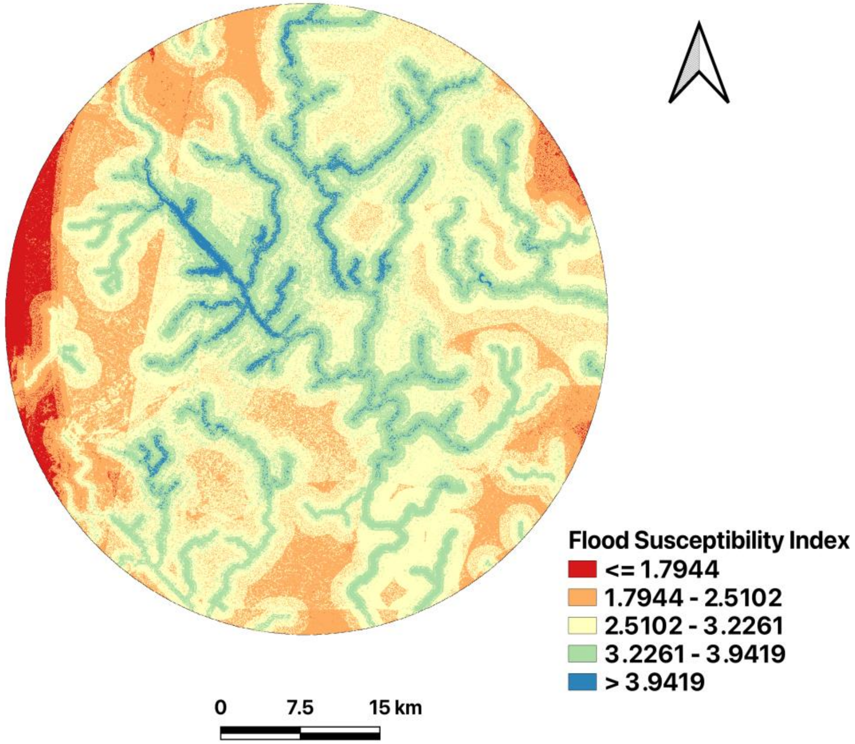


Tool for reclassifying parameters and weight assign



Tool for final flood map through weights





Final Map

- The final Flood Susceptibility Index (FSI) map classifies the study area into five susceptibility zones ranging from very low to very high flood risk.
- Blue-colored areas represent zones with very high flood susceptibility, mainly concentrated along drainage channels and low-lying regions.
- Green and light-yellow areas indicate moderate flood susceptibility, showing transitional flood-prone regions.
- Orange and red zones represent low to very low flood susceptibility, generally located in elevated or less water-accumulating areas.
- The spatial distribution of flood-prone zones highlights the strong influence of drainage patterns and topography on flood occurrence.
- This map can support flood risk management, land-use planning, and disaster mitigation strategies in the study area.

All the scripts file

https://github.com/madhubalapriya2-debug/script